

Qualitative Prime Factorization (QPF)

Gallows-Proof Methods Protocol (v1.0)

Purpose

QPF is a falsifiable methodology for discovering *empirical qualitative generators* (“primes”)—minimal sets of perceptual qualities that, together with a biologically admissible composition operator, best *predict* and *compress* perceptual data from mixture stimuli. QPF treats “primeness” as model- and constraint-relative, and operationalizes emergence via predictive generalization + complexity penalties.

1. Formal Setup

1.1 Data

Let the dataset be

$$D = \{(s_i, r_i)\}_{i=1}^N,$$

where stimuli $s_i \in \mathbb{R}_{\geq 0}^C$ are concentration vectors over C compounds and responses r_i include any subset of:

- Continuous ratings $y_i^{(c)} \in \mathbb{R}$
- Ordinal ratings $y_i^{(o)} \in \{1, \dots, K\}$
- Discrimination outcomes $y_i^{(d)} \in \{0, 1\}$ (e.g., triangle tests)
- Similarity judgments $z_{ab} \in [0, 1]$ or forced-choice comparisons

1.2 Two-Layer Structure

We separate **stimulus composition** from **percept composition**.

Stimulus layer

- Stimulus space: $\mathcal{S} = \mathbb{R}_{\geq 0}^C$
- Physical mixture operator (example for solutions): $s \otimes t := \lambda s + (1-\lambda)t, \lambda \in [0, 1]$.
- Carrier (identity stimulus): $\mathbf{1}_S := \mathbf{0}$ (base matrix/solvent; domain-specific).

Percept layer

We represent percepts through a **latent coordinate** z and a **representation map** R that yields an observable percept object $\mu \in \mathcal{P}$.

- Latent percept coordinate: $z \in \mathbb{R}^m$ (e.g., descriptor logits, intensity coordinates, or a learned embedding).
- Representation map R :
- **Option A (Simplex)**: $\mu = R(z) = \text{softmax}(z) \in \Delta^{m-1}$
- **Option B (Gaussian)**: $\mu = R(z, \Sigma) = \mathcal{N}(z, \Sigma)$

Define the percept mapping (learned):

$$z_i, \Sigma_i \sim F_{\phi}(s_i), \mu_i := R(z_i, \Sigma_i).$$

(For simplex, Σ_i is omitted.)

Induced percept composition

Percept composition is defined in latent space via $\otimes_z$ (family-dependent) and then mapped to μ :

$$z_a \otimes z_b := z_a \otimes_z z_b \quad \mapsto \quad \mu_a \otimes \mu_b := R(z_a \otimes z_b, \Sigma_{a \otimes b}).$$

Neutral percept: $\mathbf{1}_P := R(0, \Sigma_0)$ (or $\text{softmax}(0)$ for simplex).

A candidate percept composition operator $\otimes_P$ (implemented via $\otimes_z$ and R) is **admissible** if it satisfies the following (interpreted in the chosen representation):

2.1 Continuity

For all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d(\mu_a, \mu_a') < \delta \mapsto d(\mu_a \otimes \mu_b, \mu_a' \otimes \mu_b) < \epsilon.$$

2.2 Monotonicity (on designated intensity functionals)

Choose intensity functionals $I_\ell(\mu)$ (e.g., sweetness, bitterness), typically linear in latent space (e.g., $I_\ell(\mu) = w_\ell^\top z$). Require:

$$I_\ell(\mu_a) \leq I_\ell(\mu_b) \mapsto I_\ell(\mu_a \otimes \mu_c) \leq I_\ell(\mu_b \otimes \mu_c)$$

for relevant ℓ .

2.3 Contractivity (intensity magnitude)

Let $J(\mu)$ be an overall intensity magnitude (e.g., $\|z\|_2$). Require:

$$J(\mu_a \otimes \mu_b) \leq \max\{J(\mu_a), J(\mu_b)\}.$$

(This allows nonlinear interactions in *shape/direction* while preventing runaway “super-intensity.”)

2.4 Approximate associativity

$$d(\big((a \otimes b) \otimes c, a \otimes (b \otimes c)\big) < \epsilon.$$

2.5 Carrier invariance

$$\mathbf{1}_P \otimes \mu = \mu \otimes \mathbf{1}_P = \mu.$$

2.6 Enforcement as priors/penalties

Admissibility is imposed via constrained parameters and/or soft penalties:

$$p(\theta) \propto \exp\left(-\sum_k \lambda_k \mathrm{violation}_k(\theta)\right).$$

3. Computability Layer (Type-2 Effectivity / Computable Analysis)

3.1 Assumptions

- Law class $\mathcal{L} = \{L_1, L_2, \dots\}$ computably enumerable.
- Weights $\pi(L_i)$ uniformly lower semicomputable with $\sum_i \pi(L_i) \leq 1$.
- Point probabilities $(i, n) \mapsto \mathbb{P}_{L_i}(N = n)$ uniformly computable as reals.

3.2 Core results

Theorem 3.2.1 (Effectivity of count mixtures)

$$\lambda_N(n) := \sum_{i=1}^{\infty} \pi(L_i) \mathbb{P}_{L_i}(N=n)$$

is a **lower semicomputable semimeasure**.

Corollary 3.2.2 (Unattainability)

Any q that is not a lower semicomputable semimeasure cannot equal λ_N under the above assumptions.

Theorem 3.2.3 (Universality)

Every lower semicomputable semimeasure is attainable in some computable setup (allowing degenerate laws).

3.3 Structural restriction: convolution closure

Assume:

1. $(W, \otimes)$ is a commutative monoid (disjoint-union semantics).
2. Additivity: $N(w \otimes w') = N(w) + N(w')$.
3. Independent composition: for independent $w, w' \sim L$, the composed world has law $L^{\otimes 2}$.
4. Closure under k -fold replication.

Then for each L , the count distribution $p_L = (N)_* \mathbb{P}_L$ satisfies

$$p_{L^{\otimes k}} = p_L^{*k} \quad \text{for all } k \in \mathbb{N}.$$

Hence mixtures $\lambda_N = \sum_L \pi(L) p_L$ lie in the convex hull of a convolution-closed family.

3.4 Tail bound theorem (explicit tail hypotheses)

If subsystem count S and per-subsystem maximum E have exponential tails and $N_{\max} \leq \eta E$ with $\eta \leq S$, then

$$\Pr(N_{\max} \geq n) \leq C \exp(-\alpha \sqrt{n})$$

for some $C, \alpha > 0$ determined by those tail constants and the η -mixing.

4.1 Canonical gustatory pathway scaffold

1. Receptor binding (example Hill form): $R_c(s) = \frac{s_c}{K_c + s_c}$

2. Early population drive: $x_n = \sum_c w_{nc} R_c(s) + \epsilon_n$

3. Divisive normalization (canonical sensory computation):

$$y_m = \frac{\sum_n v_{mn} x_n}{\sigma_m + \sum_k \alpha_{mk} x_k}$$

1. Percept readout (example logits): $z = \beta^\top y$ and $\mu = R(z)$.

Let $\lambda \in [0, 1]$ be the physical mixture weight.

Family 1: LRS (Linear receptor-sum surrogate)

$$z_a \otimes b = \lambda z_a + (1 - \lambda) z_b.$$

Constraints: nonnegative tuning where appropriate (domain-justified).

Family 2: DN (Divisive normalization)

$$z_a \otimes b = \frac{\lambda z_a + (1 - \lambda) z_b}{\sigma + \lambda |z_a|_1 + (1 - \lambda) |z_b|_1}, \quad \sigma > 0.$$

Family 3: LB (Logistic binding / sparse interactions)

$$z_a \otimes b = \lambda z_a + (1 - \lambda) z_b + \rho \cdot (z_a \odot z_b), \quad \rho \in \text{sparse and bounded}.$$

Then $\mu_{a \otimes b} = \text{softmax}(z_{a \otimes b})$ (simplex) or $\mathcal{N}(z_{a \otimes b}, \Sigma_{a \otimes b})$ (Gaussian).

5. Statistical Model (Likelihoods + Mixture-Consistency Penalty)

5.1 Measurement likelihoods

Let μ_i be the percept object for stimulus s_i .

Continuous ratings

$$y_i \sim \mathcal{N}(I_c(\mu_i), \sigma_c^2).$$

Ordinal ratings (probit)

$$\Pr(y_i \leq k) = \Phi\left(\frac{\tau_k - I_o(\mu_i)}{\sigma_o}\right), \quad \tau_1 < \dots < \tau_{K-1}.$$

Triangle discrimination

For pair (a, b) :

$$\Pr(\text{correct}) = \gamma + (1 - \gamma) \cdot \Phi(\beta \cdot d(\mu_a, \mu_b) - \alpha), \quad \gamma = 1/3.$$

Similarity judgments

Use a bounded likelihood (recommended):

- **Beta regression:** $z_{ab} \sim \text{Beta}(\kappa m, \kappa(1 - m))$, with $m = \exp(-\lambda_s d(\mu_a, \mu_b)^2)$.

5.2 Mixture-consistency penalty (soft constraint)

For mixture-structured trials $(i, j) \in \mathcal{M}$, impose:

$$-\log p(D \mid \cdot) \leftarrow -\log p(D \mid \cdot) - \eta \sum_{(i,j) \in \mathcal{M}} \max\{0, \text{Div}(\mu_i \otimes j \mid \mu_i \otimes \mu_j) - \epsilon\},$$

where Div is KL (simplex) or W_2 (Gaussian) and the hinge avoids over-penalizing already-consistent fits.

5.3 Participant random effects

For participant u :

$$\phi_u \sim \mathcal{N}(\bar{\phi}, \Sigma_{\phi}), \quad \theta_u \sim \mathcal{N}(\bar{\theta}, \Sigma_{\theta}).$$

6. Model Comparison and Prime Inference (Anti-Gaming)

6.1 Clustered cross-validation

Primary generalization target is **novel mixtures**.

- Ratings: leave-one-stimulus-out (clustered by stimulus)
- Discrimination: leave-one-pair-out
- Pointwise PSIS-LOO reported only as sensitivity analysis

For model families $M_1, \dots, M_J$:

- Compute clustered ELPD (LOO/CV) and its SE.
- Prefer M_a over M_b if $\Delta \text{ELPD}_{ab} > 2 \cdot \text{SE}(\Delta)$.
- Otherwise use stacking/model averaging.

6.3 Stacking weights (model averaging)

Compute stacking weights w_j to maximize expected log score of the mixture predictive distribution.

6.4 Generator (prime) inference

QPF supports two complementary inference modes.

Mode A (Primary): Bayesian inclusion (spike-and-slab)

Introduce inclusion indicators $\gamma_q \in \{0, 1\}$ with spike-and-slab priors, and infer

$$P(q \in G \mid D, M_j) = \mathbb{E}[\gamma_q \mid D, M_j].$$

Define stacking-weighted robust probability:

$$P_{\text{robust}}(q) = \sum_{j=1}^J w_j P(q \in G \mid D, M_j).$$

Mode B (Secondary check): MDL / nested CV compression

Perform **nested CV** once per model family (not per posterior draw):

$$\text{Score}(G) = \text{NLL}_{\text{outerCV}}(G) + \lambda_1 |G| + \lambda_2 |\theta|_0.$$

Use the Bayes-inclusion ranking to define a manageable candidate pool (e.g., top k qualities), then search over subsets.

6.5 Structure-preserving permutation null

To test MDL-prime claims:

- Permute labels within concentration strata.
- Permute edge labels on the mixture-design graph.
- Preserve adjacency/mixture structure.

6.6 Prime definitions

- **Bayes-prime:** $P(q \in G \mid D) > 0.95$ (within a family)
- **Robust-prime:** $P_{\text{robust}}(q) > 0.90$ (stacking-weighted)
- **MDL-prime:** ΔScore_q exceeds the 95th percentile of the structure-preserving null

7. Algorithm 1 (Complete QPF Pipeline)

1. Specify $\otimes$ -families $\{\text{LRS}, \text{DN}, \text{LB}\}$ with admissibility constraints.
2. For each family M_j : fit the hierarchical model to D , including mixture-consistency penalty.
3. Compute clustered CV/ELPD and SE for each M_j ; compute stacking weights w_j .
4. For each M_j : infer inclusion probabilities $P(q \in G \mid D, M_j)$ (spike-and-slab).

5. Compute $P_{\text{robust}}(q) = \sum_j w_j P(q \in G \mid D, M_j)$.
 6. Run structure-preserving permutations; compute MDL-prime checks via nested CV.
 7. Report: winning $\otimes$ family (or model-averaged results), robust primes, model-dependent primes, and all failure diagnostics.
-

8. Falsifiable Predictions

8.1 Basic tastes (within canonical physiology-constrained families)

- {sweet, sour, salty, bitter} should have high P_{robust} (> 0.9).
- umami should require its own generator (often > 0.8).

8.2 Bittersweet emergence

- “bittersweet” should be generator-worthy primarily under DN/LB and primarily when discrimination/similarity data are included.

8.3 Category boundary

- Extending QPF to taste + trigeminal (astringent/metallic/pungent) should fail unless $\otimes$ is expanded to cross-modal interactions or the generator set includes cross-modal primitives.
-

9. Success / Failure Criteria

Supported if

1. One $\otimes$ family dominates (stacking weight > 0.7) or stacking delivers stable predictions.
2. Robust primes are consistent across participants (e.g., ICC > 0.7).
3. Held-out discrimination performance is high (e.g., $> 75\%$ or meaningfully above a baseline model).
4. Compression ratio < 0.5 (generators fewer than half of qualities).

Falsified if

1. Models are indistinguishable (stacking weights $\sim$ uniform) *and* model averaging yields no stable prime structure.
 2. No quality achieves robust-prime status.
 3. Predictive performance on novel mixtures is worse than a simple baseline.
 4. No compression: required generators $\approx$ all qualities.
-

10. Conjectural Research Direction (Regularity)

Under Lipschitz continuity in stimulus space and Wasserstein-2 similarity:

1. $|\lambda_N(n+1) - \lambda_N(n)| \leq C/n^\alpha$
 2. No knife-edges under small TV perturbations
 3. Compressibility: $K(\lambda_N[1:n]) = O(\sqrt{n} \log n)$
-

11. Implementation Notes

- Bayesian inference: Stan (NUTS/HMC), multiple chains, posterior predictive checks, and simulation-based calibration.
 - PSIS-LOO / stacked weights: use `loo` tooling, monitor Pareto $\hat{k}$ diagnostics.
 - Prefer clustered CV for dependence-heavy designs.
 - Keep all priors explicit and motivated by biology (normalization, saturation, nonnegativity where appropriate).
-

12. External References (selected)

Computable analysis / TTE / semimeasures

- Klaus Weihrauch. *Computable Analysis: An Introduction*. Springer (2000). ISBN 978-3-540-66817-6. doi: 10.1007/978-3-642-56999-9.
- Vasco Brattka, Guido Gherardi, Arno Pauly. *Weihrauch Complexity in Computable Analysis* (arXiv: 1707.03202).
- Ray Solomonoff. *The Discovery of Algorithmic Probability* (retrospective; discusses universal a priori probability).
- Scholarpedia entry: *Algorithmic probability* (overview of Solomonoff induction and universal priors).

Model evaluation and averaging

- Aki Vehtari, Andrew Gelman, Jonah Gabry. *Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC* (arXiv:1507.04544).
- Aki Vehtari et al. *Pareto Smoothed Importance Sampling* (JMLR 2024; PSIS foundations).
- Yuling Yao, Aki Vehtari, Daniel Simpson, Andrew Gelman. *Using stacking to average Bayesian predictive distributions* (Bayesian Analysis 2018; arXiv:1704.02030).

MDL / sparsity priors

- Jorma Rissanen. *Modeling by shortest data description*. Automatica (1978).

- Andrew Barron, Jorma Rissanen, Bin Yu. *The Minimum Description Length Principle in Coding and Modeling* (IEEE IT 1998).
- Edward I. George, Robert E. McCulloch. *Variable Selection Via Gibbs Sampling*. JASA (1993).
- Carlos M. Carvalho, Nicholas G. Polson, James G. Scott. *The horseshoe estimator for sparse signals*. Biometrika (2010).

Neural computation and taste mixture interactions

- Matteo Carandini, David J. Heeger. *Normalization as a canonical neural computation*. Nat Rev Neurosci (2012). doi:10.1038/nrn3136.
- David J. Heeger. *Normalization of cell responses in cat striate cortex*. Visual Neuroscience (1992).
- B. G. Green et al. *Taste mixture interactions: suppression, additivity, and the predominance of sweetness* (2010).
- Bob Carpenter et al. *Stan: A Probabilistic Programming Language*. Journal of Statistical Software (2017).

Citizen Gardens © 2025 CC-NC-ND 4.0